CSE 270 Sports Analytics

Quiz 1 Answers

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Problem 1

What is the correct way of installing a library in R?

```
A) install.packages("devtools")
B) install.packages(devtools)
C) library("devtools")
D) library(devtools)
```

Answer

```
A) install.packages("devtools")
```

Explanation

library is used to call a package that already is installed, you can use the package name with **library** function without putting it in a string in case if it is already installed, but when installing a new package you need to provide the name of the package in a string.

Problem 2

Do these two codes perform the same action ?

```
a <- c(1:10)
a <- c(1,10)

A) TRUE
B) FALSE
```

Answer

B) FALSE

Explanation

```
print(c(1:10))
```

```
##  [1]  1  2  3  4  5  6  7  8  9 10
```

```
print(c(1,10))
```

```
## [1]  1 10
```

Problem 3

a) Get the schedule for CSE 270,
Pick all the correct options.

```
A) class_schedules["CSE270"]
B) class_schedules[0]
C) class_schedules[1]
D) class_schedules$CSE_270
```

Answer

```
A) class_schedules["CSE270"]
C) class_schedules[1]
```

Explanation

C) is wrong as indexing starts from 1 in R.
D) is wrong as for vectors $ opreator is not available.

Problem 3

b) Get the schedule for all classes except ***CSE 270***,
Pick all the correct options.

```
A) class_schedules[-c("CSE270")]
B) class_schedules[2:4]
C) class_schedules[c(2,4)]
D) class_schedules[names(class_schedules) != "CSE270"]
```

Answer

```
B) class_schedules[2:4]
D) class_schedules[names(class_schedules) != "CSE270"]
```

Explanation

A) is wrong as its used for dataframes and is not working for vectors.
C) is wrong as it will select only the second and fourth classes.

Problem 4

Which of these code chunks will create a 3x3 matrix
and fill it with numbers from 1 to 9.
It should look like this.

```
A) one_to_nine <- matrix(1:9)
B) one_to_nine <- matrix(1:9, byrow=TRUE, nrow=3)
C) one_to_nine <- matrix(c(1,9), byrow=TRUE, nrow=3)
D) one_to_nine <- matrix(rbind(c(1:3), c(4:6), c(7:9)))
```

Answer

```
C) one_to_nine <- matrix(c(1,9), byrow=TRUE, nrow=3)
```

Explanation

If you do not specify the attributes **byrow** and **nrow**, the function matrix will create a 1 x n dimensional matrix where n is the number of elements in your provided vector, even if you provide a vector that is not one dimensional, like in D) .

Problem 5

a) Write a code chunk to convert the matrix ***one_to_nine*** into a dataframe and name that dataframe ***standings_2020_2022***.

Answer

a) Write down the commands to change the column names and row names of the matrix ***one_to_nine***, set the column names to ("Lazio", "Roma" , "Juventus") and the row names to ("2022", "2021", "2020")

```
colnames(one_to_nine) <- c("Lazio", "Roma" , "Juventus")
rownames(one_to_nine) <- c("2022", "2021", "2020")
```

Problem 5

b) The matrix displays the standings for the teams during the last three years, write a command to get Lazio's final position in 2020, can you get it in multiple ways ?

Answer

```
print(one_to_nine["2020","Lazio"])
```

```
## [1] 7
```

```
print(one_to_nine[3,1])
```

```
## [1] 7
```

```
print(one_to_nine["2020", 1])
```

```
## [1] 7
```

Problem 5

c) Write down the output of this code chunk

```
rowSums(one_to_nine)
```

```
## 2022 2021 2020
##    6   15   24
```

Problem 6

a) Write a code chunk to convert the matrix ***one_to_nine*** into a dataframe and name that dataframe ***standings_2020_2022***.

```
# Option A
standings_2020_2022 <- as.data.frame(one_to_nine)
# Option B
standings_2020_2022 <- data.frame(one_to_nine)
```

Problem 6

Difference between data.frame and as.data.frame

data.frame() can be used to build a data frame while as.data.frame() can only be used to coerce other object to a data frame.
for example:

```
(df1 <- data.frame((1:9),1:3))
```

```
##   X.1.9. X1.3
## 1      1    1
## 2      2    2
## 3      3    3
## 4      4    1
## 5      5    2
## 6      6    3
## 7      7    1
## 8      8    2
## 9      9    3
```

```
(df2 <- as.data.frame((1:9),1:3))
```

```
##   (1:9)
## 1     1
## 2     2
## 3     3
```

Problem 6

b) Write a code chunk to add one more column to the data frame, name it "Napoli" and add values
(4, 2, 3) for the values of that column.

```
# Option A
standings_2020_2022$Napoli <- c(4,2,3)
# Option B (Creative one :D )
Napoli <- c(4,2,3)
standings_2020_2022 <- cbind(standings_2020_2022, Napoli)
```

Problem 6

c) Write a code chunk to add one more row to the data frame, name it 2019 and add the values, (3,4,2,5)

```
standings_2020_2022['2019',] <- c(3,4,2,5)
```

Problem 7

a) R has a built in dataframe called ***iris*** that contains some information about Species. Write down the commands to get the data frame and show the last 3 observations of the data frame.

```
data(iris)
tail(iris,3)
```

```
##     Sepal.Length Sepal.Width Petal.Length Petal.Width   Species
## 148          6.5         3.0          5.2         2.0 virginica
## 149          6.2         3.4          5.4         2.3 virginica
## 150          5.9         3.0          5.1         1.8 virginica
```

b) Write a code chunk to create 2 new variables for the dataframe iris and call them ***Sepal.Area*** and ***Petal.Area*** and fill in the values with the area(width x length) of both variables correspondingly,

```
iris$Sepal.Area <- iris$Sepal.Length * iris$Sepal.Width
iris$Petal.Area <- iris$Petal.Length * iris$Petal.Width
```

Problem 7

c) Write a code chunk to filter out the data frame ***iris*** to only include the species for which the variable ***Sepal.Area*** is greater than 10 and the variable ***Petal.Area*** is less than 1.

```
iris[iris$Sepal.Area > 10 & iris$Petal.Area < 1,]
```

```
##    Sepal.Length Sepal.Width Petal.Length Petal.Width Species Sepal.Area
## 1           5.1         3.5          1.4         0.2  setosa      17.85
## 2           4.9         3.0          1.4         0.2  setosa      14.70
## 3           4.7         3.2          1.3         0.2  setosa      15.04
## 4           4.6         3.1          1.5         0.2  setosa      14.26
## 5           5.0         3.6          1.4         0.2  setosa      18.00
## 6           5.4         3.9          1.7         0.4  setosa      21.06
## 7           4.6         3.4          1.4         0.3  setosa      15.64
## 8           5.0         3.4          1.5         0.2  setosa      17.00
## 9           4.4         2.9          1.4         0.2  setosa      12.76
## 10          4.9         3.1          1.5         0.1  setosa      15.19
## 11          5.4         3.7          1.5         0.2  setosa      19.98
## 12          4.8         3.4          1.6         0.2  setosa      16.32
## 13          4.8         3.0          1.4         0.1  setosa      14.40
## 14          4.3         3.0          1.1         0.1  setosa      12.90
## 15          5.8         4.0          1.2         0.2  setosa      23.20
## 16          5.7         4.4          1.5         0.4  setosa      25.08
## 17          5.4         3.9          1.3         0.4  setosa      21.06
## 18          5.1         3.5          1.4         0.3  setosa      17.85
```

Problem 7

c)

Using dplyr

```
dplyr::filter(iris,(Sepal.Area > 10 & Petal.Area < 1))
```

```
##    Sepal.Length Sepal.Width Petal.Length Petal.Width Species Sepal.Area
## 1           5.1         3.5          1.4         0.2  setosa      17.85
## 2           4.9         3.0          1.4         0.2  setosa      14.70
## 3           4.7         3.2          1.3         0.2  setosa      15.04
## 4           4.6         3.1          1.5         0.2  setosa      14.26
## 5           5.0         3.6          1.4         0.2  setosa      18.00
## 6           5.4         3.9          1.7         0.4  setosa      21.06
## 7           4.6         3.4          1.4         0.3  setosa      15.64
## 8           5.0         3.4          1.5         0.2  setosa      17.00
## 9           4.4         2.9          1.4         0.2  setosa      12.76
## 10          4.9         3.1          1.5         0.1  setosa      15.19
## 11          5.4         3.7          1.5         0.2  setosa      19.98
## 12          4.8         3.4          1.6         0.2  setosa      16.32
## 13          4.8         3.0          1.4         0.1  setosa      14.40
## 14          4.3         3.0          1.1         0.1  setosa      12.90
## 15          5.8         4.0          1.2         0.2  setosa      23.20
## 16          5.7         4.4          1.5         0.4  setosa      25.08
## 17          5.4         3.9          1.3         0.4  setosa      21.06
## 18          5.1         3.5          1.4         0.3  setosa      17.85
```